

Error Decomposition

$$E_{\text{total}} = E_{\text{grounding}} + E_{\text{geom}}$$

illustrative query: distance(door, table)

$$E_{\text{geom}} = 0.000 \text{ m}$$

surface engine is exact

$$E_{\text{grounding}} = 0.833 \text{ m}$$

VLM picked the wrong instance

Centroid baseline (model-independent)

$$d_{\text{cent}} - d_{\text{surf}} \approx 1.0 \text{ m}$$

constant bias, removed only by surface distance